MATH 101 Unit 4: Exponentials, Logarithms, and Systems - Practice 2

Version 202610122020.44 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Solve both equations exactly. Check the logarithmic domain in (b).

(*a*) $4^{x-2} = 2^{x+3}$

(*b*) $\log_3(x-2) + \log_3(4) = 2$

2. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10.

(*a*) $\log(0.4) + \log(2500)$

(*b*) $\frac{\log_4(27)}{\log_4(9)}$

3. (a) Solve the linear system. (b) Solve the nonlinear system algebraically. (c) Test whether (0,0) belongs to the region given by the inequalities; show both comparisons.

(*a*) $x + y = 5$

$x - y = 1$

(*b*) $y = x^2$

$y = x + 2$

(*c*) $y \geq x^2,\ y < x + 2$

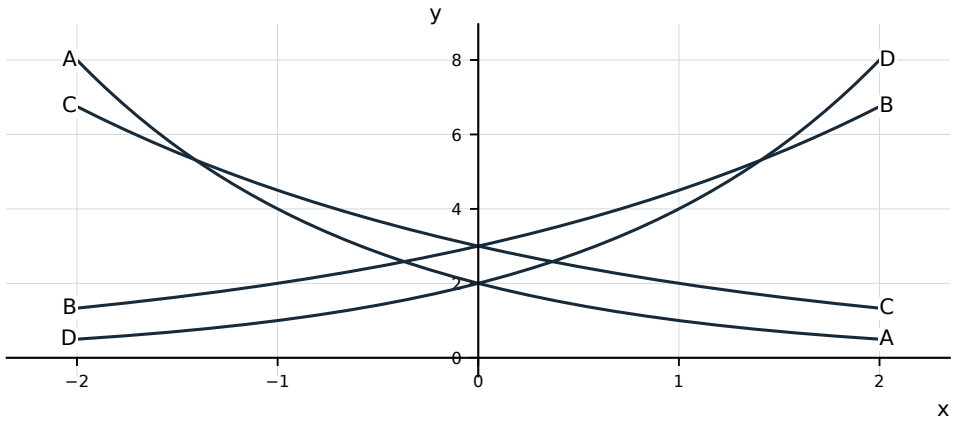
4. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $3(\frac{2}{3})^x$

(*b*) $2(\frac{1}{2})^x$

(*c*) $2(2)^x$

(*d*) $3(\frac{3}{2})^x$



5. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain.

$f(x) = \log_3(x - 2) + 2$

Student answer sheet

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Question 1

(a)

8 pts

(b), with domain

12 pts

Question 2

(a)

10 pts

(b)

10 pts

Question 3

(a) Ordered pair

6 pts

(b) Intersections

10 pts

(c) Membership

4 pts

Student answer sheet

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Question 4

(a)

5 pts

(b)

5 pts

(c)

5 pts

(d)

5 pts

Question 5

(a) Domain

4 pts

(a) Range

4 pts

(b) Asymptote

3 pts

(b) x-intercept

3 pts

(c) Left end

3 pts

(c) Right end

3 pts

Answer key

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Question 1

(a)	$x = 7$	8 pts
(b), with domain	$x = \frac{17}{4};\ x > 2$	12 pts

Question 2

(a)	3	10 pts
(b)	$\frac{3}{2}$	10 pts

Question 3

(a) Ordered pair	$(3,\ 2)$	6 pts
(b) Intersections	$(-1,\ 1);\ (2,\ 4)$	10 pts
(c) Membership	At (0,0): 0 >= 0 and 0 < 2; yes, in the region.	4 pts

Answer key

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Question 4

(a)	C	5 pts
(b)	A	5 pts
(c)	D	5 pts
(d)	B	5 pts

Question 5

(a) Domain	$(2, \infty)$	4 pts
(a) Range	$(-\infty, \infty)$	4 pts
(b) Asymptote	$x = 2$	3 pts
(b) x-intercept	$(\frac{19}{9},\ 0)$	3 pts
(c) Left end	$x \to 2^+ :\ f(x) \to -\infty$	3 pts
(c) Right end	$x \to \infty :\ f(x) \to \infty$	3 pts